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EXPERIMENT NO. : 7 (ASSIGNMENT 3)

DATE: 18/9/2026

1. AIM & OBJECTIVES

1. To study and analyze Local Oxidation of Silicon (LOCOS) and Shallow Trench Isolation (STI) techniques using Semulator3D.

2. To study the effect of isolation thickness on device performance, determine the optimal active area, and measure optimal gate area and isolation oxide thickness (Assignment 3.1.a).

3. To study the relationship between planned gate pitch and gate length, determine the smallest achievable gate pitch, and report the corresponding gate length (Assignment 3.1.b).

2. TOOL

Coventor Semulator3D (3D Process Modeling and Virtual Fabrication Software)

3. QUESTION 1: LOCOS

Isolation in a device is required for providing electrical isolation, minimizing leakage current, and offering structural support. It is a fundamental building block that ensures the smooth operation of complex electronic devices.

This assignment explores transistor design by investigating two key isolation techniques: LOCOS and STI. We analyze the impact of oxide thicknesses and gate pitch on LOCOS performance.

The first task for this assignment is to identify isolation oxide, gate oxide, gate length, gate pitch and bird's beak during the LOCOS process. Also, we analyze the change in achieved gate length with respect to planned gate length by plotting characteristic graphs.

Local Oxidation of Silicon (LOCOS) is a traditional technique to create isolation regions between transistors and other devices on a silicon wafer. It essentially grows a thick oxide layer in specific areas of the silicon wafer, electrically isolating them from the surrounding silicon.

Silicon nitride acts like an oxidation mask, preventing oxide from forming on covered areas. A thin layer of oxide, called the pad oxide (15 nm), is added between the silicon and nitride to prevent mechanical stress during the process.

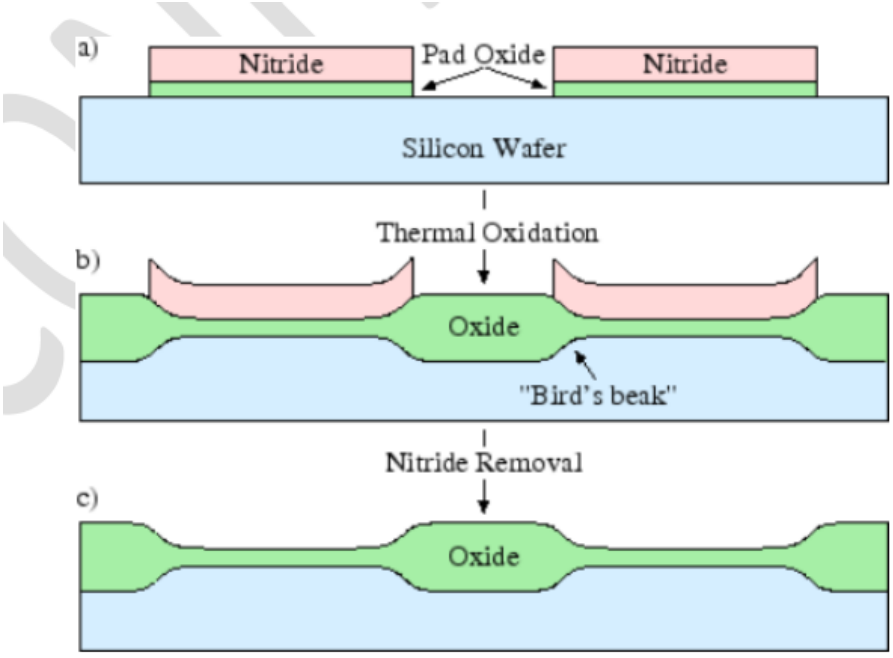
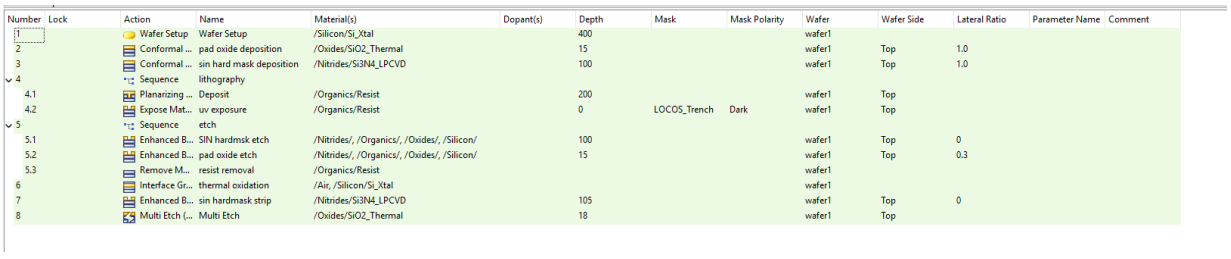


Figure 1 : LOCOS Process Flow

4. STEPS INVOLVED IN LOCOS PROCESS (PROCESS EDITOR)

Table: Steps involved in LOCOS Process (Process Editor Recipe)

Step	Action	Step Name	Material(s)	Depth (nm)	Mask	Mask Polarity
1	Wafer Setup	Wafer Setup	/Silicon/Si_Xtal	400	-	-
2	Conformal Deposit	pad oxide deposition	/Oxides/Silicon_Oxide_Thermal	15	-	-
3	Conformal Deposit	sin hard mask deposition	/Nitrides/Silicon_Nitride_LPCVD	100	-	-
4	Sequence	lithography	-	-	-	-
4.1	Planarizing Deposit	Deposit	/Organics/Resist	200	-	-
4.2	Expose Material	uv exposure	/Organics/Resist	0	LOCOS_Trench	Dark
5	Sequence	etch	-	-	-	-
5.1	Enhanced Basic Etch	SiN hardmask etch	/Nitrides/, /Organics/, /Oxides/	100	-	-
5.2	Enhanced Basic Etch	pad oxide etch	/Nitrides/, /Organics/, /Oxides/	15	-	-
5.3	Remove Materials	resist removal	/Organics/Resist	-	-	-
6	Interface Growth	thermal oxidation	/Air, /Silicon/Si_Xtal	-	-	-
7	Enhanced Basic Etch	sin hardmask strip	/Nitrides/Silicon_Nitride_LPCVD	105	-	-
8	Multi Etch	Multi Etch	/Oxides/Silicon_Oxide_Thermal	18	-	-



Number	Lock	Action	Name	Material(s)	Dopant(s)	Depth	Mask	Mask Polarity	Wafer	Wafer Side	Lateral Ratio	Parameter Name	Comment
1		Wafer Setup	Wafer Setup	/Silicon/Si_Xtal		400			wafer1				
2		Conformal ...	pad oxide deposition	/Oxides/SiO2_Thermal		15			wafer1	Top	1.0		
3		Conformal ...	sin hard mask deposition	/Nitrides/Si3N4_LPCVD		100			wafer1	Top	1.0		
4		Sequence	lithography										
4.1		Planarizing ...	Deposit	/Organics/Resist		200			wafer1	Top			
4.2		Expose Mat...	uv exposure	/Organics/Resist		0	LOCOS_Trench	Dark	wafer1	Top			
5		Sequence	etch										
5.1		Enhanced B...	SiN hardmsk etch	/Nitrides/, /Organics/, /Oxides/, /Silicon/		100			wafer1	Top	0		
5.2		Enhanced B...	pad oxide etch	/Nitrides/, /Organics/, /Oxides/, /Silicon/		15			wafer1	Top	0.3		
5.3		Remove M...	resist removal	/Organics/Resist					wafer1				
6		Interface Gr...	thermal oxidation	/Air, /Silicon/Si_Xtal					wafer1				
7		Enhanced B...	sin hardmask strip	/Nitrides/Si3N4_LPCVD		105			wafer1	Top	0		
8		Multi Etch (...	Multi Etch	/Oxides/SiO2_Thermal		18			wafer1	Top			

Figure 2 : Steps involved in LOCOS Process (Process Editor)

5. SIMULATED LOCOS PROCESS STEPS (3D & 2D VIEWS)

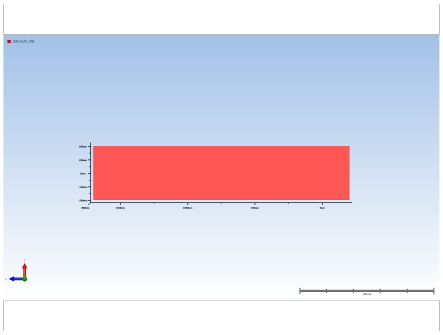


Figure 3 : 1. Wafer Setup

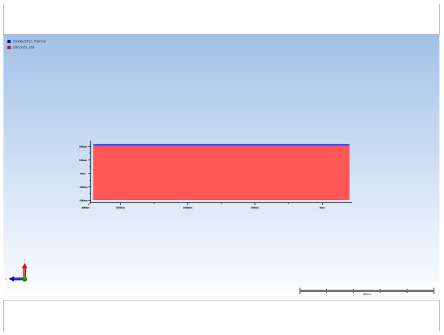


Figure 4 : 2. Pad Oxide Deposition

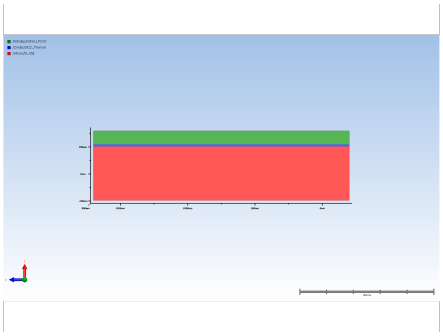


Figure 5 : 3. Silicon Nitride Deposition

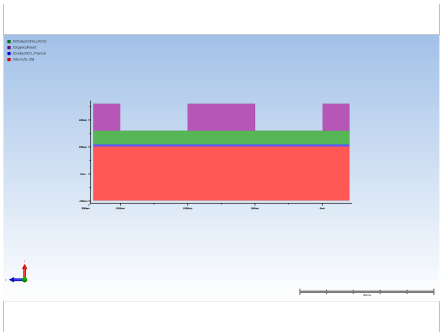


Figure 6 : 4. Lithography

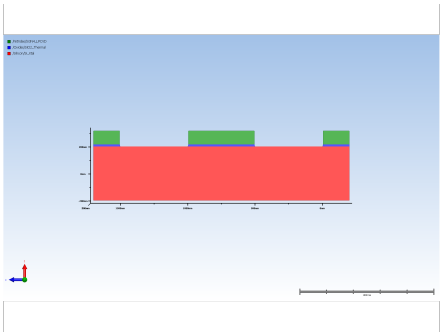


Figure 7 : 5. Etch

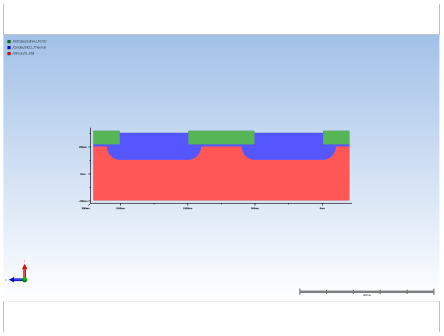


Figure 8 : 6. Thermal Oxidation

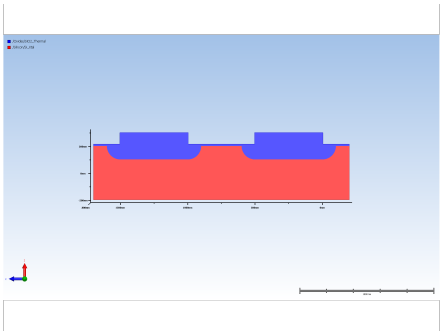


Figure 9 : 7. Nitride Strip

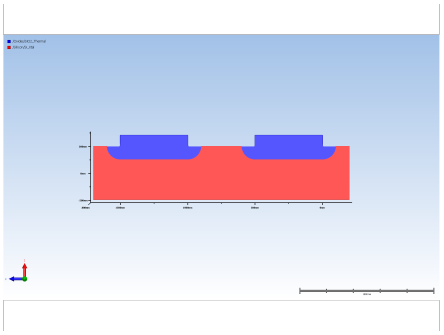


Figure 10 : 8. Multi Etch (Final LOCOS Structure)

6. THICKNESS SPECIFICATION & DEVICE DIMENSIONS

Gate oxide is pad oxide deposited before the nitride layer on the region above the active area of the device, while isolation oxide is a thick layer of thermally grown oxide which provides insulation between adjacent devices.

Table 1 defines the desired nominal specifications compared against the measured dimensions extracted from the Semulator3D simulation model.

Table 1 : Thickness Specification

S.No.	Parameter	Desired Value (nm)	Measured Value (nm)
01.	Wafer	400	398.0
02.	Gate Oxide (Pad Oxide)	15	15.0
03.	Isolation Oxide	100	180.0
04.	Gate Length	300	300.34
05.	Gate Pitch	350	494.59
06.	Isolation Length	100	505.41

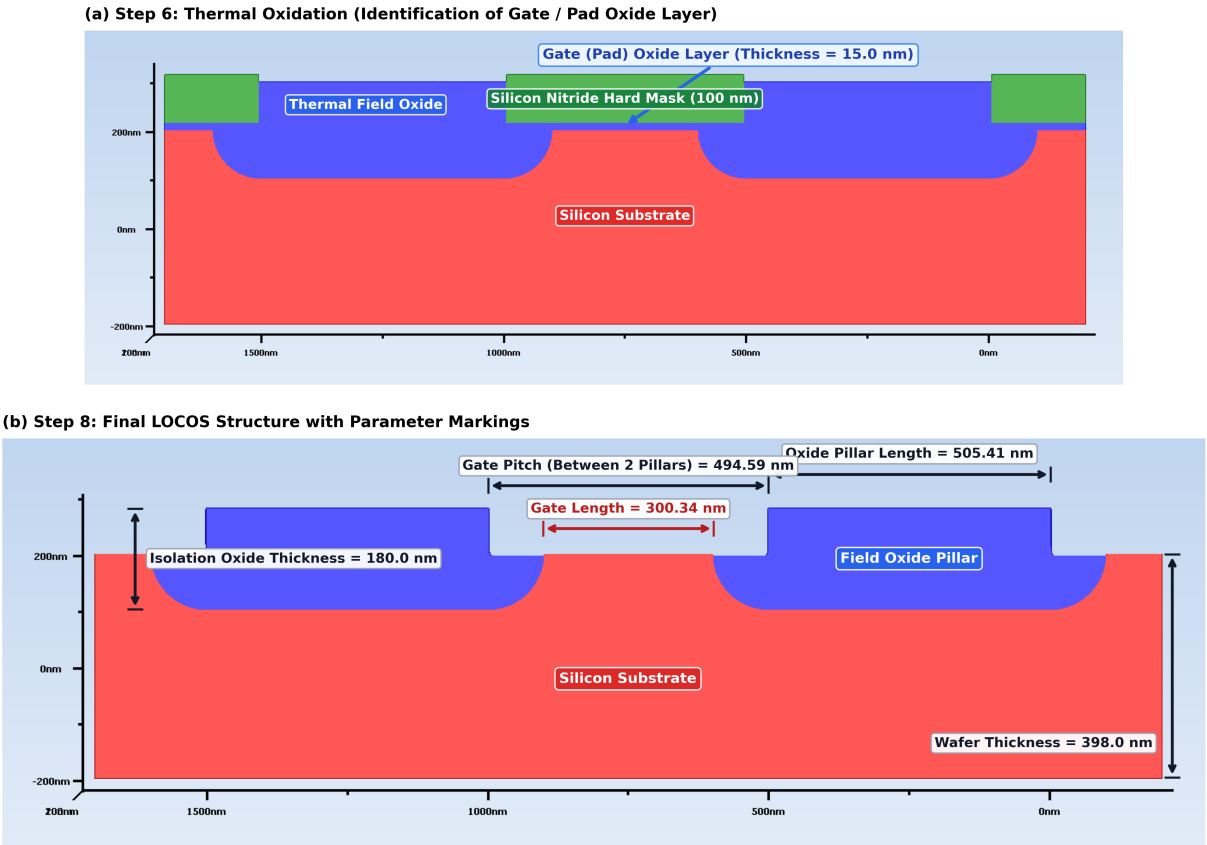


Figure 11 : Simulated Gate and Isolation Oxide Dimensions

7. 1. QUESTION 1(A) : OPTIMAL ACTIVE AREA

LOCOS takes advantage of the different oxidation rates of silicon and silicon nitride to create isolated regions on a silicon wafer. Silicon nitride acts like an oxidation mask, preventing oxide from forming on covered areas. A thin layer of oxide, called the pad oxide, is added between the silicon and nitride to prevent mechanical stress during thermal oxidation.

The final product is an insulating layer (isolation oxide) surrounding the desired active regions. However, due to the pad oxide, lateral oxidant diffusion occurs under the nitride mask edge, forming a tapered oxide wedge known as the bird's beak.

Limitations of LOCOS:

1. Bird's Beak Formation: Lateral oxidant diffusion through the pad oxide lifts the nitride layer from the corners, consuming silicon beneath the active mask.

2. Scalability Problem: As device geometries scale to smaller dimensions, lateral oxidation severely encroaches into the active area, causing the electrical gate length to become significantly less than the drawn mask dimension.

To overcome these limitations, thermal oxidation can be performed at higher temperatures (where oxidation rate increases faster than lateral diffusion), or modern manufacturing transitions to Shallow Trench Isolation (STI) which replaces thermal field oxidation with etched vertical trenches filled with deposited oxide and planarized by chemical mechanical polishing.

This section deals with finding the optimal active area, that is, where field oxide thickness provides the required dielectric isolation without excessive active gate length encroachment. Active gate length depends on several critical layout and process parameters, including planned gate pitch, isolation oxide thickness, and isolation length.

Table 2 : Case for finding optimal active area

S.No.	Desired Isolation Oxide Thickness (nm)	Measured Isolation Depth (nm)	Measured Gate Length (nm)	Measured Isolation Length (nm)
1	100	80.0	400.84	505.51
2	120	100.0	380.74	505.51
3	150	130.0	350.72	505.51
4	180	160.0	320.72	505.51
5	200	180.0	300.72	505.51
6	300	280.0	200.72	505.51
7	320	300.0	180.72	505.51

8. ASSIGNMENT 3.1.(A) : EFFECT OF ISOLATION THICKNESS

Study the effect of isolation thickness on device performance:

1. Simulation Procedure: Simulations were conducted across desired isolation oxide thicknesses from 100 nm to 320 nm (Table 2).

2. Gate Length Variation Trend: As measured isolation depth increases from 80.0 nm to 300.0 nm, measured active gate length shrinks monotonically from 400.84 nm down to 180.72 nm. This active channel reduction is directly caused by lateral oxidant diffusion lifting the nitride hardmask edges.

3. Optimal Gate Area and Isolation Oxide Thickness: The optimal isolation oxide thickness is determined to be 180.0 nm (nominal recipe target 200 nm), yielding an active gate length of 300.34 nm and an isolation length of 505.41 nm. At this operating point, robust inter-device electrical isolation is achieved while maintaining the planned 300 nm active channel dimensions.

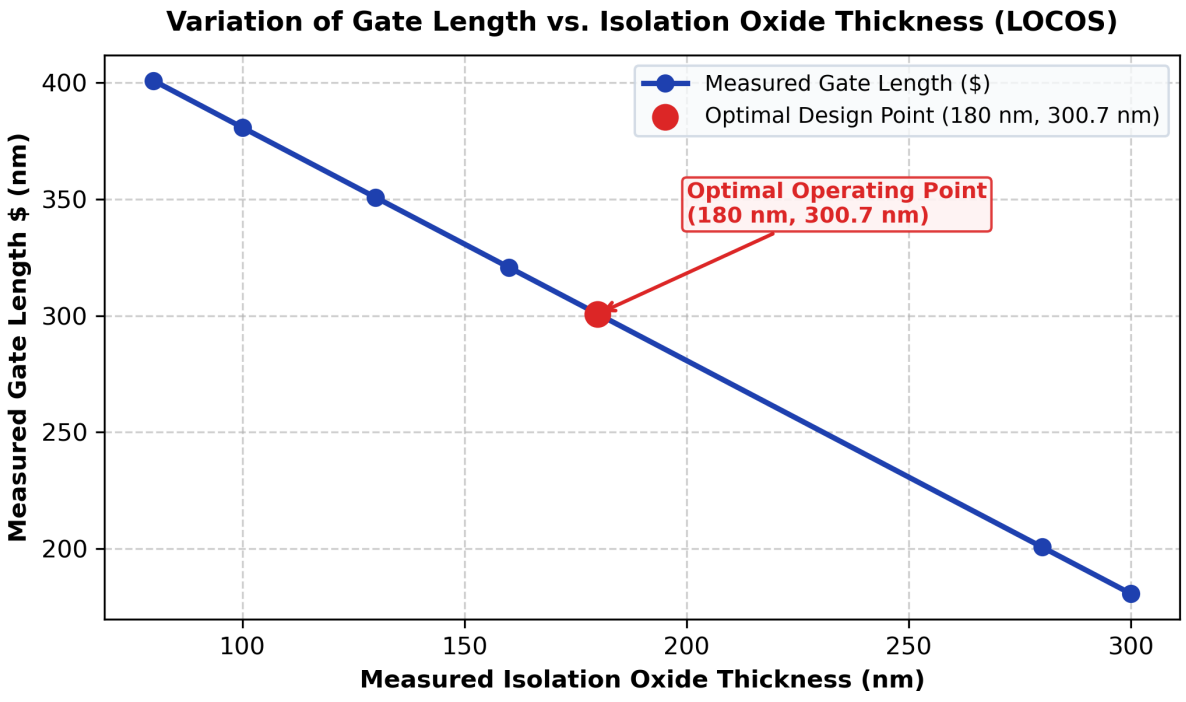


Figure 12 : Variation of Gate Length vs. Isolation Oxide Thickness

9. 2. QUESTION 1(B) / ASSIGNMENT 3.1.(B) : PLANNED GATE PITCH VS. GATE LENGTH

In this part, we analyze the variation in gate length with changes in mask layout. The distance between two trenches (planned gate pitch) in layout is varied from 350 nm down to 150 nm, as shown in Table 3.

• Isolation oxide thickness entered = 100 nm.

Table 3 : Cases for Variation of Planned Gate pitch vs gate length

S.No.	Planned Gate Pitch (nm)	Measured Isolation Depth (nm)	Measured Gate Length (nm)	Measured Isolation Length (nm)
1	350	80.0	250.84	505.51
2	300	80.0	200.84	505.51
3	250	80.0	150.84	505.51
4	200	80.0	100.84	505.51
5	150	80.0	50.84	505.51

Study the relationship between planned gate pitch and gate length:

1. Scaling Relationship: Across planned gate pitches from 350 nm to 150 nm, the achieved gate length scales linearly according to Planned Gate Pitch = Gate Length + 93.65 nm. The constant offset of ~93.65 nm represents the necessary process margin between active gates.

2. Smallest Achievable Gate Pitch: The smallest achievable gate pitch determined from the analysis is 144.49 nm, which corresponds to a gate length of 50.84 nm.

3. Physical Scaling Limit: Below 50 nm gate length, opposing bird's beak encroachment fronts merge, resulting in complete pinch-off and loss of active silicon area.

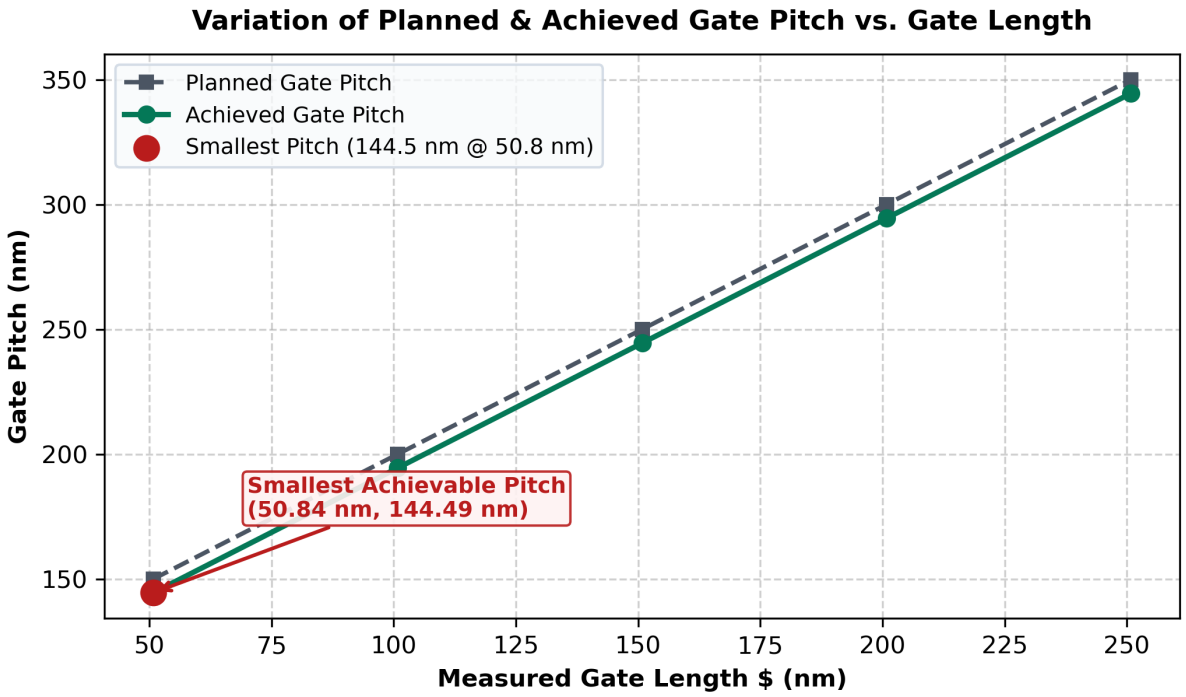


Figure 13 : Variation of Planned Gate Pitch vs. Gate Length

10. OBSERVATIONS & DISCUSSION

1. Stress Relief and Lateral Diffusion Trade-Off: The 15 nm pad oxide layer successfully relieves interfacial mechanical stress between silicon and silicon nitride. However, it concurrently acts as a lateral diffusion channel for oxidant species during high-temperature field oxidation.

2. Isolation Oxide vs. Active Dimension Trade-Off: Increasing isolation thickness enhances dielectric isolation and inter-device breakdown voltage, but severely reduces effective active gate length from 400.84 nm to 180.72 nm. Optimal baseline balance is achieved at 180.0 nm isolation depth with 300.34 nm gate length.

3. Pitch Scaling Boundary: Gate pitch scales linearly with gate length down to 144.49 nm at a gate length of 50.84 nm. Below this physical limit, lateral oxidation encroachment destroys device functionality.

11. CONCLUSION

The Local Oxidation of Silicon (LOCOS) fabrication sequence was successfully modeled and analyzed using Coventor Semulator3D. The 8-step virtual fabrication demonstrated selective thermal oxidation and lateral encroachment formation.

Quantitative parameter extraction validated that thicker field oxide reduces active channel length due to lateral oxide encroachment. The optimal operating point was identified as 180.0 nm isolation oxide thickness and 300.34 nm gate length.

Pitch variation confirmed that the smallest achievable gate pitch is 144.49 nm at a 50.84 nm gate length. These findings clearly demonstrate the fundamental scaling limitations of LOCOS and highlight why advanced semiconductor technology nodes adopt Shallow Trench Isolation (STI) for dense integrated circuits.